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Recommendations for the organization of mental health services for children and adolescents in Belgium: Use of the soft systems methodology

Philippe Vandenbroeck[a], Rachel Dechenne[a], Kim Becher[a], Marijke Eyssen[b], Koen Van den Heede[b,*]

[a] shiftN, Belgium
[b] Belgian Healthcare Knowledge Centre, Belgium

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ABSTRACT

The prevalence of mental health problems among children and adolescents in Western countries is high. Belgium, like many other Western countries, struggles with the set-up of a coherent and effective strategy for dealing with this complex societal problem. This paper describes the development of a policy scenario for the organization of child and adolescent mental health care services (CAMHS) in Belgium. The development process relied on Soft Systems Methodology including a participatory process with 66 stakeholders and a review of the existing (inter-)national evidence. A diagnostic analysis illustrated that the Belgian CAMHS is a system in serious trouble characterized by fragmentation and compartmentalization. A set of 10 strategic recommendations was formulated to lay down the contours of a future, more effective CAMHS system. They focus on mastering the demands made on scarce and expensive specialized mental health services; strengthening the range of services – in particular for those with serious, complex and multiple mental health problems – and strengthening the adaptive capacity of and the ethical guidance within the future CAMHS system.

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1. Introduction

The prevalence of mental health problems and disorders among children and adolescents in Western countries is high. Indeed, an overall prevalence rate of 14% for all clinically important child and adolescent mental health disorders in Western countries was found by Waddell et al., while the prevalence of co-morbidity of two or more mental disorders varied from 47% to 68% [1]. These conclusions were based on a review of large population-based studies including interview-based diagnostic and functional evaluations. Precise population-based epidemiological data is lacking for Belgium. For some child and adolescent mental health services, figures are available on how many children and adolescents are taken in care, but for other services, e.g. private consultations with child psychiatrists or outpatient clinics, such figures are lacking [2].

In 2005, the World Health Organization (WHO) called for national action in the field of child and adolescent mental health services (CAMHS) [3]. This call for a dedicated approach was motivated by high prevalence rates of mental health problems in this age group, and by the impediment of these problems on their development and their potential impact on ongoing distress and disability in adulthood [4,5]. The WHO recommended that national action does not only cover classic child psychiatry but also the so-called "adjacent sectors" like welfare, justice, disability care, education [3].

* Corresponding author at: Belgian Health Care Knowledge Centre (KCE), 55 Boulevard du Jardin Botanique, 1000 Brussels, Belgium. Tel.: +32 2 287 33 23.
E-mail address: Koen.vandenheede@kce.fgov.be (K. Van den Heede).

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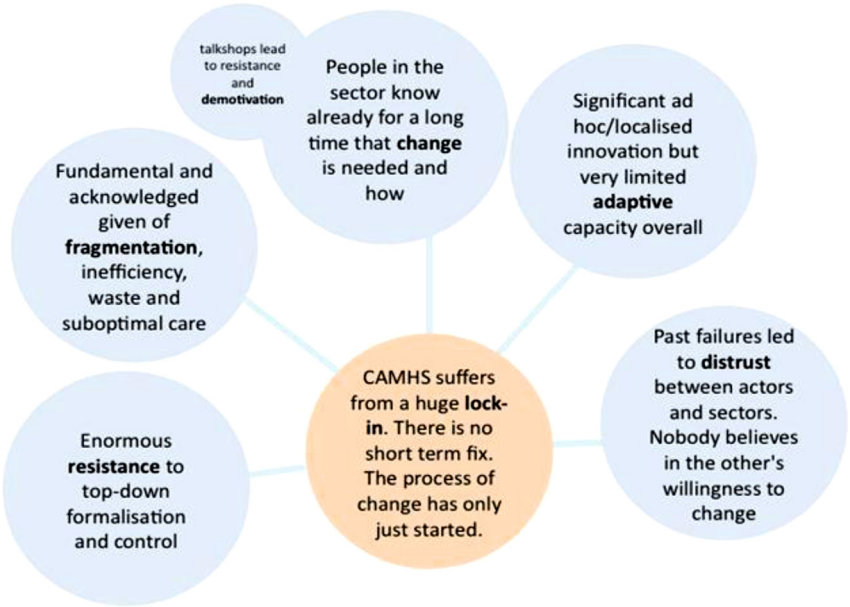

Fig. 1. The CAMHS system: a wicked problem.

Belgium, however, does not have a clear-cut CAMHS strategy. Reforms in the past have, as in other Western countries, mainly focused on the adult mental health sector, which traditionally has been characterized by large isolated institutions. Since the beginning of the 21st century, the "balanced care" model is gradually gaining influence in this sector [6,7]. This model implies that care is offered and delivered as close as possible to the patient's living environment, and only if necessary in an institution.

The CAMHS-sector developed much later and should include a developmental perspective. Indeed, the consecutive developmental stages of children and adolescents have an impact on vulnerability to disorders, and on treatment options. Therefore, a specific approach might be needed.

The Belgian Healthcare Knowledge Centre (KCE) was commissioned by the Belgian Ministry of Public Health to perform a study (June 2011–March 2012) that would offer input for a reform of the CAMHS-system. The objective of this study is to set up a participatory process to lay down the contours of a future Belgian CAMHS-system.

2. Methods

The approach used in this study is based on "Soft Systems Methodology" [8]. Soft Systems Methodology has been developed to deal with different perceptions or worldviews surrounding complex socio-technical systems. It allows those different worldviews to surface and organizes the intervention in the system under study as a learning process in which these different appreciations are confronted with one another. Therefore this methodology fits hands in gloves with the reform of the CAMHS-system. Indeed, it is a complex system in need of a systemic reform in which stakeholders actively participate in a well-structured way [8].

For participatory process the input of stakeholders were solicited. In total 66 different stakeholders (including *professionals* from different professional groups of the CAMHS sector and related sectors such as education, juvenile justice, disability care, welfare; *NGO representatives* such as patient/family right advocates; *administrators and policy makers*) participated at one or more of 5 stakeholder events (in-depth interviews, workshops and discussion tables). Their input has been complemented with information from a narrative review on the effectiveness of different organizational models in CAMHS [2], a content analysis of CAMHS policies in 3 other countries/regions (i.e., The Netherlands, The United Kingdom, British Colombia, Canada) [2], by building on previous KCE-studies on organizational aspects of mental health care for children (especially aspects of networking between different levels of care [9]; emergency psychiatric care for children and adolescents [10] as well as on insights from systems thinking methodology [8,11–17].

Both the results of the participatory process and the existing documents were used to make a diagnostic analysis of the existing CAMHS system and to identify solution elements for a future, more effective system.

Two sets of outputs were created based on these diagnostic analyses and solution elements:

- The *root definition* (purpose) and *activity model* (or conceptual model of required activities), delineating the ideal contours of the future CAMHS system, but without specifying organizational boundaries, legal status, resources, and so on.
- A set of general *strategic policy recommendations*, based on a comparison of the current Belgian CAMHS-system

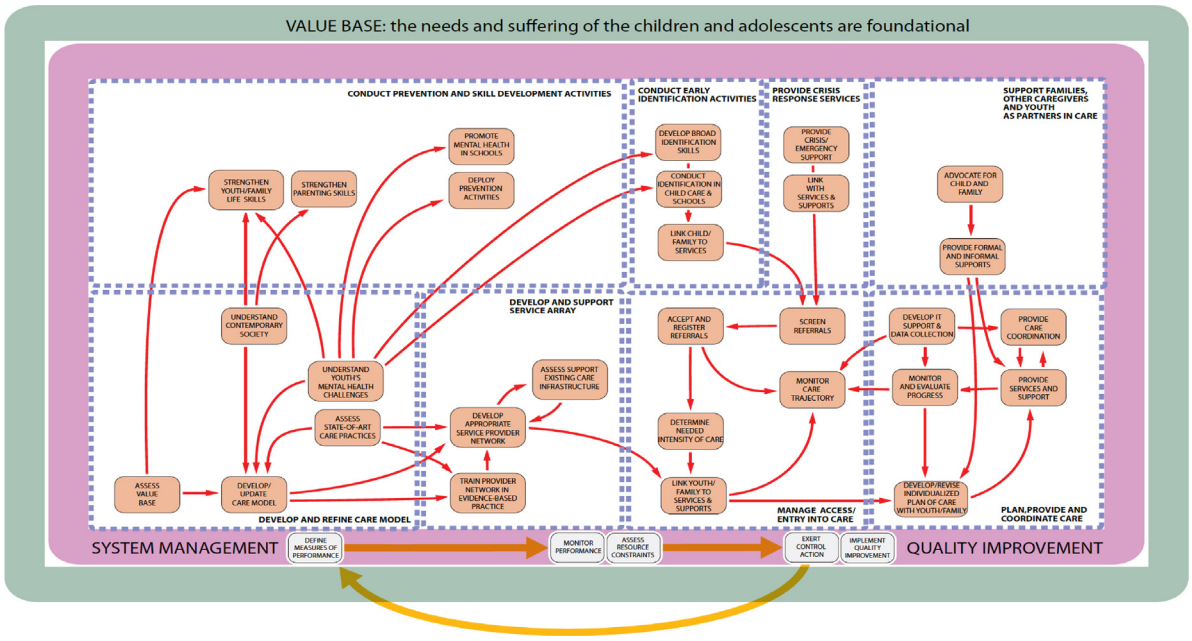

Fig. 2. Activity model.

(i.e. *stakeholder mapping*) with this idealized system to identify overlaps and gaps (i.e. *gap analysis*).

A draft version of the recommendations was submitted to stakeholders in a consolidation workshop and amended based on their insights and contributions.

3. Results

3.1. The Belgian CAMHS-system 'locked in'

There was a broad consensus among the stakeholders on the problematic state of affairs in the CAMHS sector. *Several problem areas* were identified, such as long waiting lists, an ever increasing demand, the absence of a clear overarching vision and of a transparent evaluation framework. The problems are clearly *interconnected* and the system suffers from inefficiency, suboptimal care and fragmentation, driven by ideological and sectoral diversity between the different services, and by a lack of collaborative perspectives and skills of the professionals. A systems model was developed to visualize the interdependency between these challenges (see Fig. 1). Notwithstanding this, the sector is also characterized by a lot of goodwill and commitment, by expertise, and by pockets of innovation and efficiency.

The need for change has been debated extensively in the past and is well recognized by the sector. While many *innovative local initiatives have emerged bottom-up* in the past years, the CAMHS *system as a whole failed to adapt*. This institutional 'lock in' has led to distrust between actors and sectors and to a suspicion as regards other parties' willingness to change. Yet, stakeholders voiced a strong *resistance to top-down formalization and control*. Consequently, there is a duality between the strong appreciation of stakeholders for a bottom-up approach and the difficulty to set up effective organizational and management practices to develop interagency collaboration.

All elements taken together, we conclude that reform of the CAMHS system can be considered as a so-called wicked problem [18]. Typical for 'wicked problems' is that they manifest a high degree of technical and social complexity. Bottlenecks in the CAMHS system are numerous and interconnected and there are important differences in perception amongst the many stakeholders about the right way forward to improve the effectiveness and efficiency of the system. Hence, obvious *short-term fixes for the many challenges burdening the Belgian CAMHS-system are unlikely to exist*.

3.2. Purpose and required activities of a future CAMHS system

The *root definition* that resulted from an analysis of the diagnostic and solution elements describes an idealized future system as follows:

> The CAMHS system is an array of ethically guided, professionally supported and evidence-based services for children and adolescents with, or at risk of, mental health challenges and their entourage, that are provided in a coordinated, personalized, developmentally appropriate, and culturally competent manner in the least restrictive environment that is clinically appropriate and best adapted to the child's needs, to help these young people to increase their wellbeing and potential for development, at home, in school, in the community and throughout life.

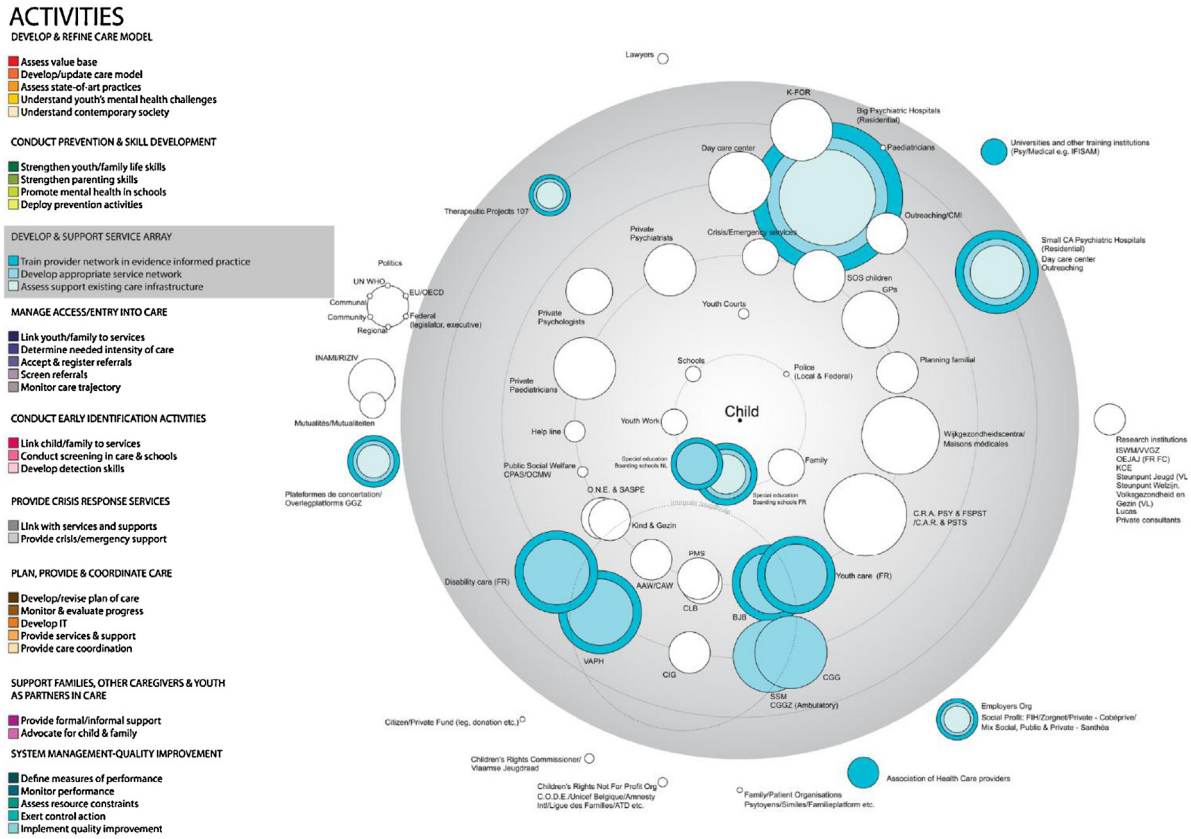

Fig. 3. Stakeholder map with additional coding of activities.

The activity model (see Fig. 2) shows in a generic way the 35 activities to be carried out to realize the system's purpose, grouped in 9 functional modules:

- plan, provide and coordinate care;
- manage access/entry into care;
- provide crisis/emergency response services;
- develop and support service array;
- support families, other caregivers and youth as partners in care;
- conduct prevention and life skill development activities;
- conduct early identification activities;
- develop and refine care model;
- system management and quality improvement.

The activity model is a vehicle to support *three basic transformations*: at the *family and child level* the CAMHS system needs to assure that the suffering of the child is mitigated; at the *services level* it needs to support better co-ordination; at the *policy level* it has to ensure adaptiveness and fidelity to the value base. The activity model is built with the aim to depict what is necessary to obtain a coordinated care model to which several care systems and actors can contribute.

Since the activity model is *generic*, it allows for different organizational architectures when operationalizing it.

The activity model was complemented by a *stakeholder mapping* (see Fig. 3 for an example). For each functional module of the model a stakeholder map was developed that showed the actors in the existing system who contributed to the activities in that module. The mapping illustrated the operational and regulatory complexity and multidimensionality of the CAMHS system but also the potential of many actors to contribute to the diverse activities that make the CAMHS system tick.

3.3. *From the envisaged CAMHS-model toward strategic recommendations*

Ten recommendations have been proposed to support a transition from the current, fragmented and siloed CAMHS system to an effective, coordinated and ethically-guided delivery of supports and care that crosses sectorial and program boundaries. These recommendations have been based on:

- The structure of the activity model of an idealized CAMHS system and the confrontation with the present.
- Solution elements that emerged from stakeholder encounters at various stages in the participatory process.
- Solution elements from the literature and existing documents (KCE-reports).

The recommendations have been organized as follows:

- The first four recommendations collectively want *to curb the demands made on scarce and expensive specialized mental health services* for children and adolescents. The aim is to ensure that young people receive appropriate care in the least restrictive and most adapted environment. This is expected to lead to a more prudent use of specialized care and residential facilities. The recommendations want to achieve this via prevention, empowerment of users and their entourage (families), adequate filtering at first line care services by supporting professional competencies in non-specialized mental health care, and by strengthening the system's capacity to deal with mental health-related crises. All of these recommendations reflect the demands imposed by the activity model of an idealized CAMHS system.
- Recommendations 5 and 6 focus on *strengthening the range of services offered by mental health professionals, particularly regarding the care for children who suffer from serious, complex and multiple mental health problems*. These complex situations may or may not be compounded by violent behavior or cultural barriers. The recommendations want to plug gaps in accountability (preventing that professionals and organizations keep on going with "passing on the buck") in dealing with these young people, by reinforcing a flexible and assertive approach to providing care for these children in their natural environment, and by improving the cultural competences of care providers.
- Recommendations 7–9 are targeted toward *strengthening the adaptive capacity of and the ethical guidance within the future CAMHS system*. They seek to do this by making sure that actors in the system, including representatives of children and families (entourage), engage in cross-sectoral forums to stimulate partnerships on an ongoing basis. It is proposed that all stakeholders commit to developing a shared vision and an ethical charter for the broader CAMHS system. It is also recommended that continuous efforts are done to better understand and visualize the needs for and supply of mental health services. Finally, there is a recommendation for evaluation methods that reinforce professionalism, quality and multidisciplinarity. These recommendations are intended to particularly support the ongoing adjustment of the care model, system management and improvement modules in the activity model.
- Recommendation 10 suggests *a couple of steps to be undertaken next*. Indeed the strategic recommendations formulated in this study only provide the general contours that will need to be operationalized by policy makers, administrators and health professionals. The study does not address the actual range of specific initiatives that should be undertaken to realize the reform. The implementation of this study has been initiated by a decision of the Interministerial conference (i.e. platform where all regional and national Ministers responsible for Public Health discuss topics that related both to national as well as regional authorities) of June 2102 to install expert groups to prepare a national policy and plan for action for CAMHS.

4. Conclusions

This participatory research project (for more details: Vandenbroeck et al. [19]) has given stakeholders the opportunity to reflect on the direction which a reform of the CAMHS system should take.

A diagnostic analysis has shown that the CAMHS system struggles with a cluster of interdependent problems, the core issue being the extreme fragmentation and compartmentalization between organizations, sectors and professions. The cost of this fragmentation is significant, both in terms of human suffering and in terms of wasted resources. Stakeholders acknowledge this state of affairs.

Over the last decade several initiatives have been taken, but isolated service level innovations, however well intentioned and brilliantly executed, may at some point add to the system's fragmentation. Manifestly, they have not been able to bolster the adaptive capacity of the system as a whole. Past failures have resulted in distrust between actors and sectors.

Stakeholders showed strong resistance to top-down formalization and control. The upshot is that professionals realize that a more joined-up and networked way of working is necessary but that there is little trust in formalizing these networks and creating new functions such as network managers and coordinators.

All these points lead to a lock-in for which a quick fix cannot be expected. The *process of change is likely to be a lengthy one*, and this research represents only a modest step along that road. The interministerial conference for Health (i.e. composed of members of the federal, community and regional governments competent for Health, it is the place where the co-operation agreements are worked out and where various procedures for collaboration and exchange of information between the different power levels are implemented) recognized the sense of urgency and the need for the development of a long-term, phased implementation plan. They commissioned a working group with the next step toward implementation and operationalization which is expected to be ready by the end of 2013. Different stakeholders – health care professional representatives; representatives of national health insurance, and care institutions; and policymakers – have convened in a working sessions. The objective of these groups is to formulate concrete proposals (i.e. long-term sustainable reform in combination with pilot-testing) that are coherent with the recommendations made in this research.

This introduces a new way of policy making in the CAMHS-system. In contrast to piecemeal or top-down solutions, often disproportionally inspired by interest groups, the need for *changes at the system level* is recognized. Admittedly, system changes are vital to support the transition to a more effective CAMHS system. On the other hand, it is important to continue to encourage local experimentation and nurture successes, particularly when they come about through intra-sector or inter-sector partnerships. The knowledge and relational capital harvested through these collaborations is the fuel on which a system-wide transformation can thrive. Bottom-up and top-down dynamics need to reinforce each other. The cross-sectoral platforms

of management and accountability are a nodal point where these two dynamics can mesh and reinforce one another through the development of strategic plans, interagency agreements, legislative proposals, funding arrangements and appropriate monitoring and evaluation protocols.

Conflict of interest

The authors declare that they have no conflict of interest.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.healthpol.2013.07.009.

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